

Tunneling Lab Walkthrough

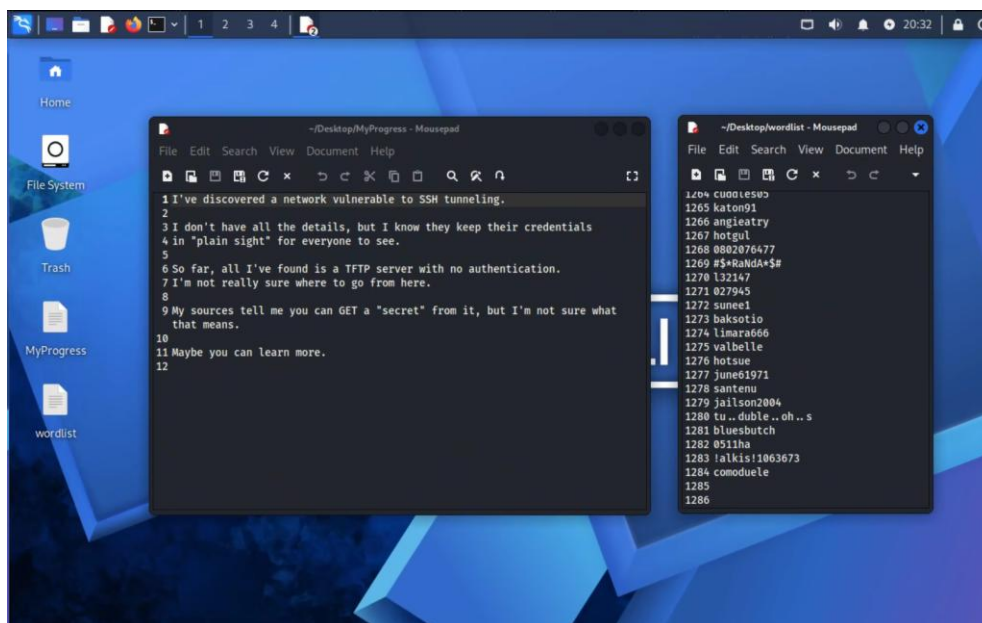
Hayden Weber

This lab is comprised of seven total virtual machines. The setup and network configuration for all of these systems is covered in-depth within the “Lab Setup Report.docx” file. This report strictly covers the process of completing this lab once it has been fully configured.

Starting Off

This lab begins on the Kali Linux *testbox* machine. The entire challenge can be completed from this box, so direct access to any of the other systems through VMware is considered cheating.

On the Kali desktop, two files are visible. These documents are named “MyProgress” and “wordlist”. The MyProgress file contains a message from someone previously attempting to attack this network. It outlines a discovered TFTP server where you can find a “secret”. The wordlist file contains over 1000 individual strings that appear to be various passwords. The file name and contents suggest this is a wordlist for cracking passwords.



Accessing TFTP

With the hint of gaining access to a TFTP server, all that's left is to find where that server is. Opening up the terminal and executing the `ip a` command will return the IP configuration for this machine's network interfaces.

```
kali@testbox: ~  
Session Actions Edit View Help  
(kali@testbox)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:ba:80:9e brd ff:ff:ff:ff:ff:ff  
    inet 192.168.19.135/24 brd 192.168.19.255 scope global dynamic noprefixroute eth0  
        valid_lft 1743sec preferred_lft 1743sec  
    inet6 fe80::eef4:bbf1:29da:b987/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:ba:80:a8 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.222.130/24 brd 192.168.222.255 scope global dynamic noprefixroute eth1  
        valid_lft 1743sec preferred_lft 1743sec  
    inet6 fe80::5997:f74e:3625:f06/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever
```

The `eth0` interface on this machine is NAT, so the address to pay attention to is on `eth1`. Next, using the `netdiscover` command on the discovered network will reveal any other active machines within it.

```
(kali@testbox)-[~]  
$ sudo netdiscover -i eth1 -r 192.168.222.0/24  
[sudo] password for kali: 
```

This outputs two addresses on the 192.168.222.0 network.

Currently scanning: Finished!		Screen View: Unique Hosts		
2 Captured ARP Req/Rep packets, from 2 hosts.		Total size: 120		
IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.222.128	00:0c:29:0f:73:27	1	60	VMware, Inc.
192.168.222.254	00:50:56:e8:24:a1	1	60	VMware, Inc.

The .254 address usually gets assigned to VMware virtual switches, so it can be assumed that the .128 address is the target machine. Regardless, this will be validated by enumerating the machine and seeing what's running on it.

The next logical step is to conduct an NMAP scan of the discovered target machine. Using the -sV option will return the current version of any discovered services.

```
(kali@testbox)-[~]
$ nmap -sV 192.168.222.128
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-24 20:50 -0500
Nmap scan report for 192.168.222.128
Host is up (0.00011s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.2p1 Ubuntu 2ubuntu3 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.66 ((Ubuntu))
MAC Address: 00:0C:29:0F:73:27 (VMware)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.97 seconds
```

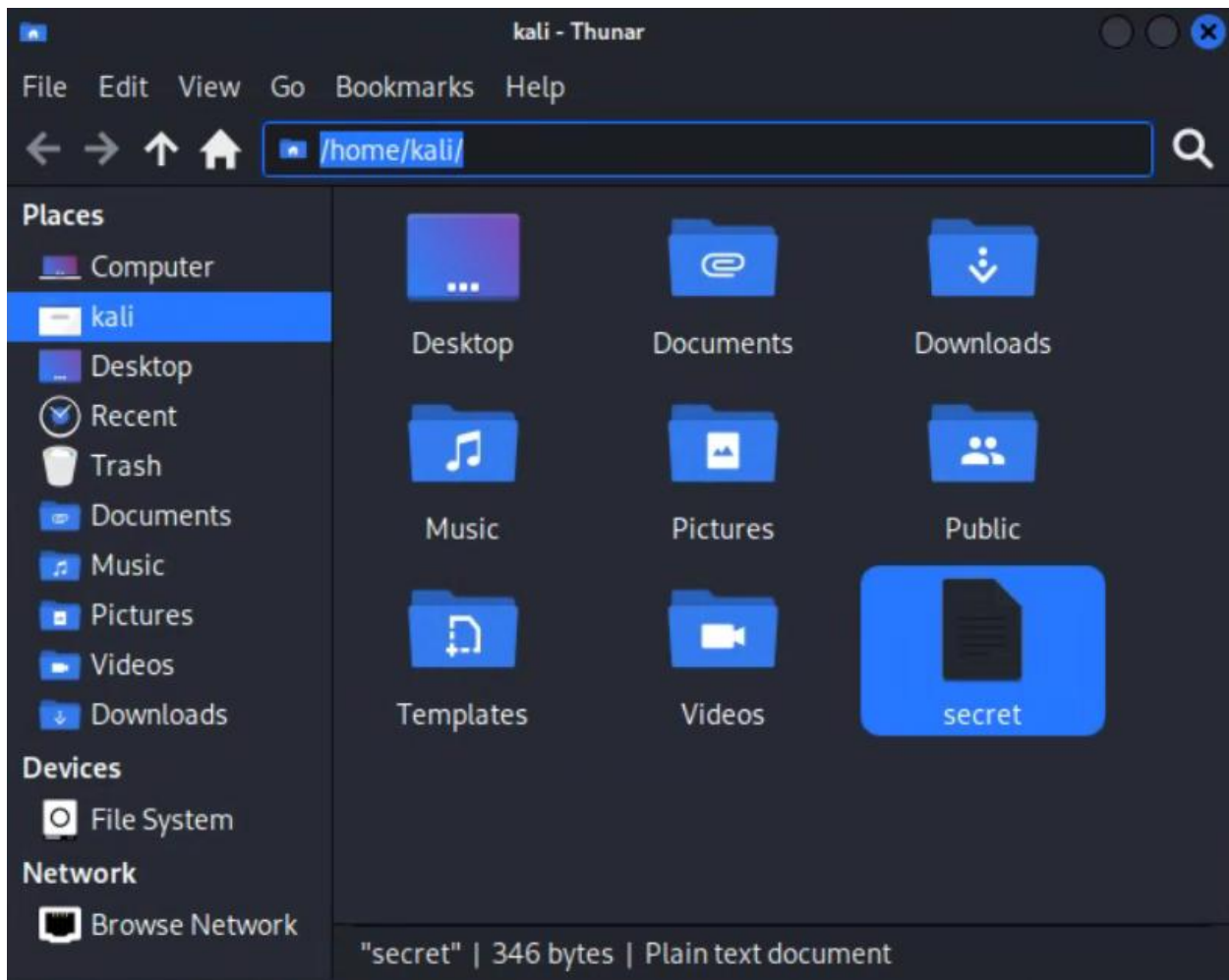
It's now confirmed that this is in fact the target system. The MyProgress document made it clear that there is a TFTP server on this target machine that requires no authentication, so it would make sense to try and access it. This can be accomplished with the tftp command.

```
(kali@testbox)-[~]
$ tftp 192.168.222.128
tftp> 
```

This logs directly into the TFTP server, no authentication needed. The MyProgress document also mentioned that someone could “GET a secret” from this server. FTP uses a get command to pull files down, so it would make sense if there was a secret file that could be downloaded from the server. This can be accomplished by using the get command followed by the file name “secret”.

```
(kali@testbox)-[~]
$ tftp 192.168.222.128
tftp> get secret
tftp> 
```

The lack of feedback means this command ran successfully. An invalid command or a request for a non-existent file would return some sort of error. Navigating to the home directory of the Kali user reveals the downloaded “secret” file.

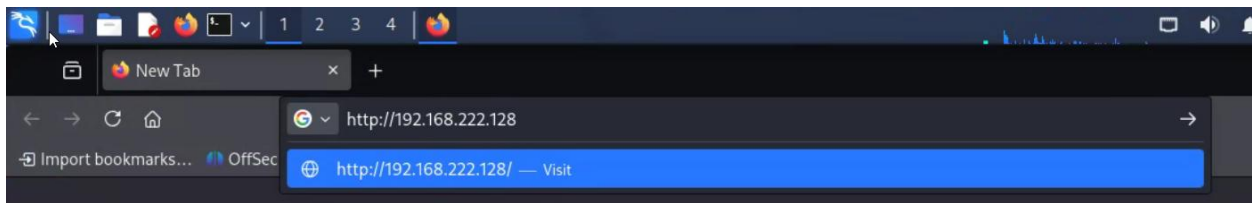


Opening this file reveals a riddle about numbers. The riddle specifically mentions that the number 80 is better than the number 69.

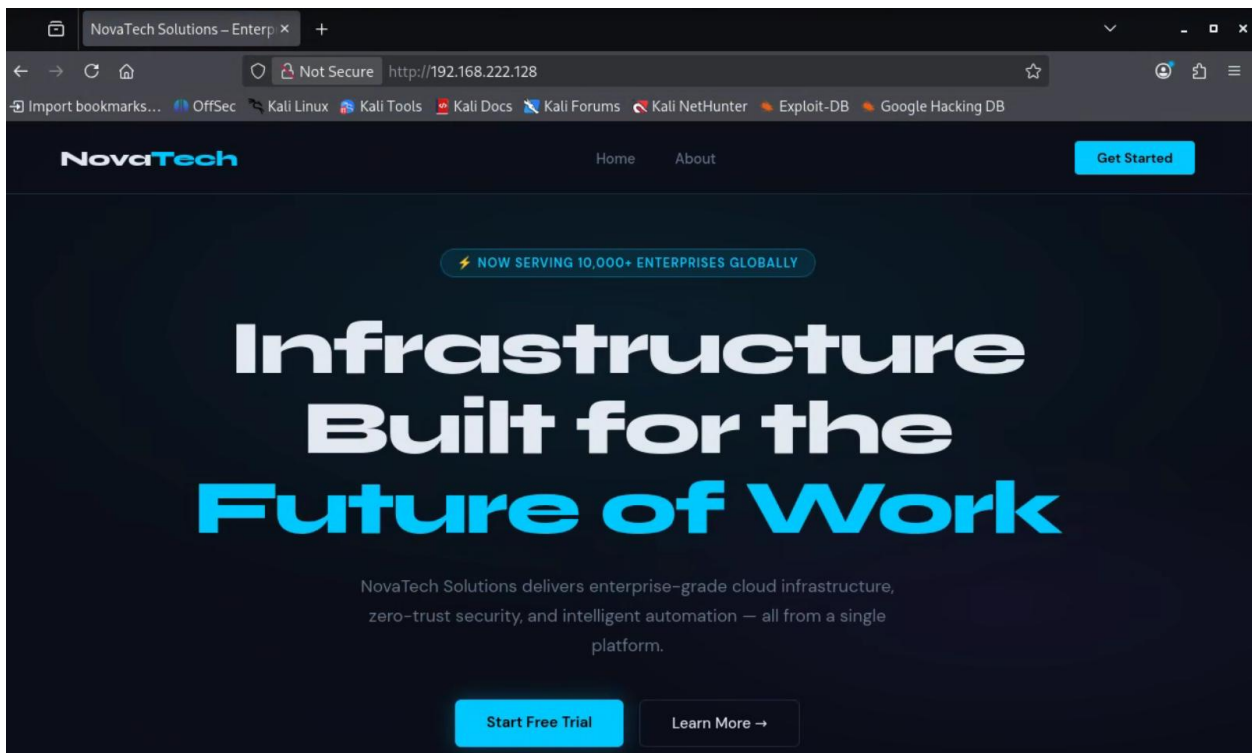


Accessing Apache Web Server

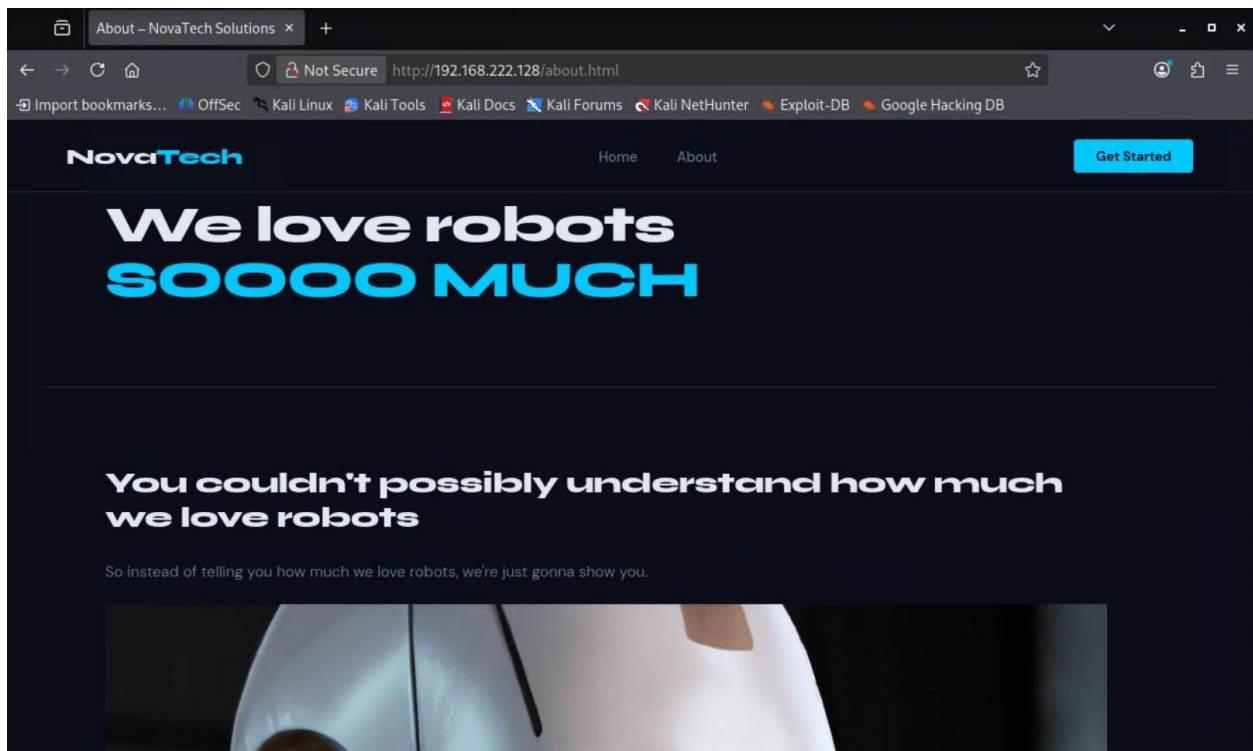
Looking at the provided riddle, the numbers likely refer to port numbers. TFTP, the service just previously accessed, uses UDP port 69. TCP port 80 is reserved for HTTP. The previous NMAP scan revealed an Apache web server on port 80, so there's likely something hidden on there. The simplest way to view this is through visiting the server IP in the Firefox browser on the Kali machine.



Visiting this site reveals a webpage for NovaTech, describing the company as a service that delivers enterprise-grade cloud infrastructure. Inspection of the home page of this website does not reveal any information that seems particularly insecure.



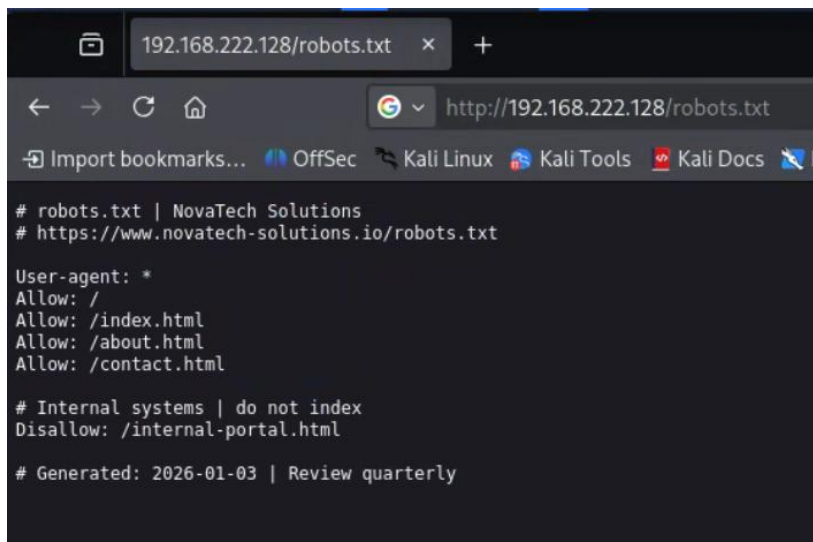
Moving on to the about section, this entire page is dedicated to robots. Specifically, how much NovaTech loves robots. Considering the website setting, this is likely a hint pointing towards the “robots.txt” file. This is a text file that instructs web crawlers to visit or not visit certain pages on a website.



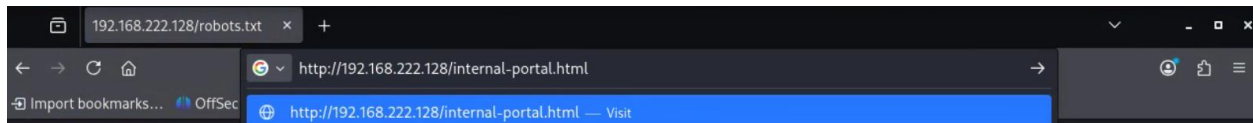
The robots.txt file can be accessed simply by replacing “/about.html” in the address bar with “/robots.txt”.



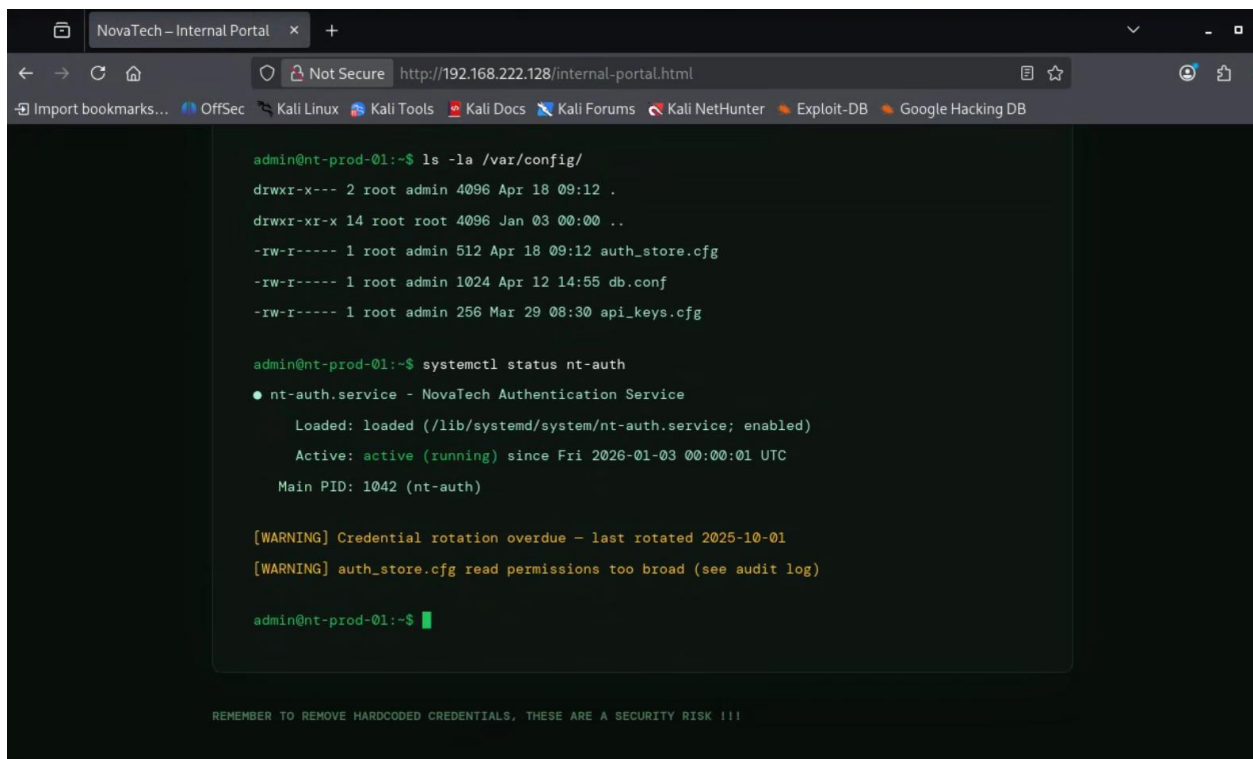
Navigating to the robots.txt file on this web server reveals that the page “/internal-portal.html” has been disallowed.



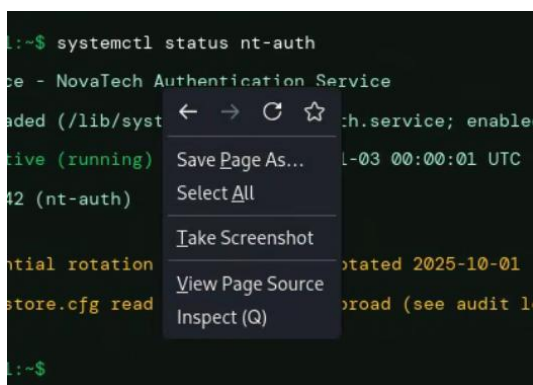
Again, this page can be accessed simply by replacing robots.txt in the address bar with the newly discovered page.



Accessing this page reveals an internal bash terminal. At the very bottom of the page, under the terminal, there is a reminder in all bold to “remove hardcoded credentials”. This suggests that credentials have been hardcoded somewhere on this page, likely in the html.



Viewing the HTML code for this page is as simple as right clicking anywhere on it and selecting “view page source”.



Inspecting the HTML code for this webpage reveals a commented-out section titled “SSH Credentials”.

```
46 </style>
47 </head>
48 <body>
49
50 <!--
51 ~~~~ NOVATECH INTERNAL SYSTEMS ~~~~
52 auth_store v2.1 | DO NOT CACHE | INTERNAL USE ONLY
53
54 [ssh-credentials]
55   arthur:2aa7bde7159defd25e2bc1a4cd5f4b51e13d5579
56   dutch:fdc01cfa3b97b2e6820be41a9f4a0d0a29dca007
57   trelawney:c8b347a32facbe02c2688c304b18e03eb25fcddf
58   john:ebf777d266a403175d74af2bc5f34ff5872666e8
59   morgan:f045e5e1b7195bc19bfd3026c044d93bb5048018
60   micah:0cc2497792b0fb6344dd03f0dd8fa27a759304cf
61 -->
62
63 <div class="terminal">
64   <div class="terminal-bar">
65     <div class="dot dot-r"></div>
```

These would appear to be usernames and password hashes for SSH on the network. Cracking these hashes would provide all the credentials necessary to tunnel through this entire network.

Further inspection of the website reveals a comment of seemingly random upper and lower case characters at the bottom of the page.

```
99
100 <div class="meta">
101   <p><b>REMEMBER TO REMOVE HARDCODED CREDENTIALS, THESE
102   </div>
103
104 <!-- dGhlIGNyZWRLbnRpYWxzIGFyZSBzaGEx -->
105 </body>
106 </html>
```

Considering the combination of both upper and lower case letters, it's likely this message has been encoded in base64. This can be decoded using the base64 command in Linux.


```
(kali@testbox)-[~]  
$ echo 'dGhlIGNyZWRLbnRpYWxzIGFyZSBzaGEx' | base64 -d  
the credentials are sha1
```

This reveals the hidden message, “the credentials are sha1”.

Password Cracking

With access to the hashed passwords and the hashing algorithm that was used on them, all that’s left is to crack them. This can be accomplished with hashcat.

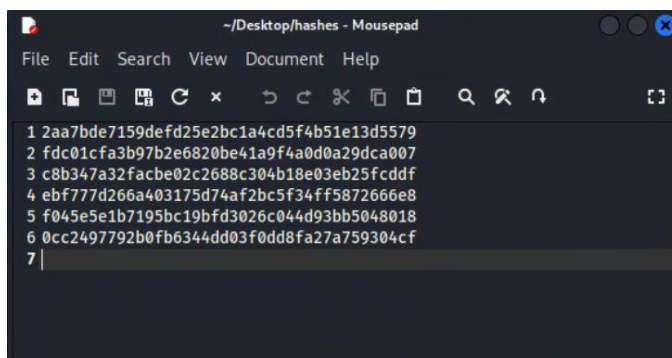
First, the hashcat mode for sha1 needs to be identified. This can be found with the man command.

```
(kali@testbox)-[~]  
$ man hashcat | grep SHA1  
100 = SHA1  
150 = HMAC-SHA1 (key = $pass)  
160 = HMAC-SHA1 (key = $salt)  
4500 = Double SHA1  
5400 = IKE-PSK SHA1  
7300 = IPMI2 RAKP HMAC-SHA1  
11200 = MySQL Challenge-Response Authentication (SHA1)
```

This reveals that mode 100 is SHA1.

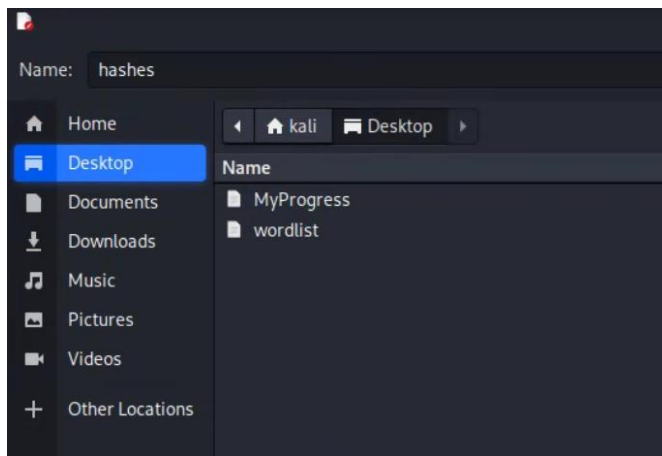
The credentials for this lab are salted with either a prefix or suffix which can include 3 symbols or digits. The fastest way to crack for all of these at once is by using hashcat rules and storing all the hashes in a single file.

First, copy all the hashes into a text file without the usernames.

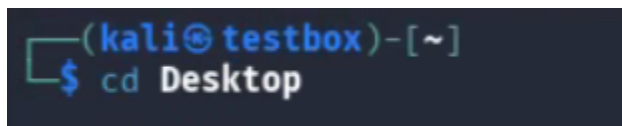


```
~/Desktop/hashes - Mousepad  
File Edit Search View Document Help  
1 2aa7bde7159defd25e2bc1a4cd5f4b51e13d5579  
2 fdc01cfa3b97b2e6820be41a9f4a0d0a29dca007  
3 c8b347a32facbe02c2688c304b18e03eb25fcdde  
4 ebf777d266a403175d74af2bc5f34ff5872666e8  
5 f045e5e1b7195bc19bfd3026c044d93bb5048018  
6 0cc2497792b0fb6344dd03f0dd8fa27a759304cf  
7 |
```

This file can be stored on the desktop as “hashes” for easy access.

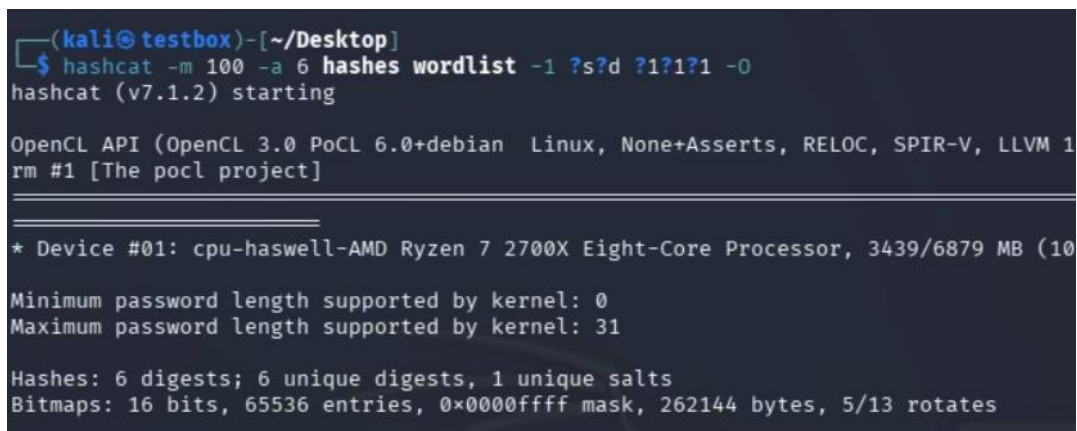


Next step is to run commands with rules to crack these hashes. Use the cd command to change to the desktop so the wordlist and hashes files can be referenced by name.

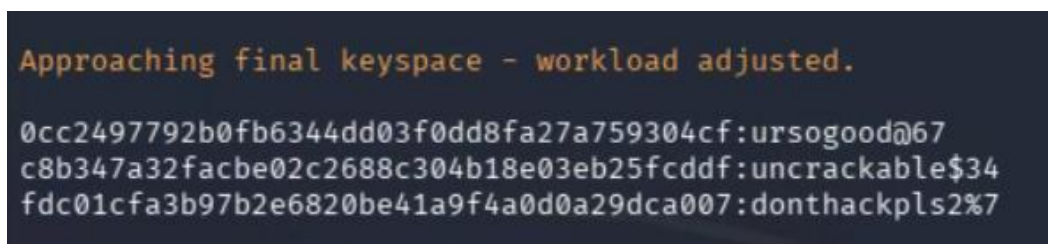


The first rule will be for suffix. (-a 6)

hashcat -m 100 -a 6 hashes wordlist -1 ?s?d ?1?1?1 -O



This returns three passwords that were successfully cracked.



These can be matched with their usernames by comparing the hashes, which makes sense to do once all the hashes have been found.

The next rule will be for prefix. (-a 7)

hashcat -m 100 -a 7 hashes -1 ?s?d ?1?1?1 wordlist -O

```
(kali@testbox)-[~/Desktop]
$ hashcat -m 100 -a 7 hashes -1 ?s?d ?1?1?1 wordlist -O
hashcat (v7.1.2) starting

OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPIR-V, LLVM #1 [The pocl project])

=====
* Device #01: cpu-haswell-AMD Ryzen 7 2700X Eight-Core Processor, 3439/6879 MB

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 31

Hashes: 6 digests; 6 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
```

This returns another three passwords that were successfully cracked, totaling the original six discovered on the webpage.

```
Approaching final keyspace - workload adjusted.

f045e5e1b7195bc19bfd3026c044d93bb5048018:$98hackerman
ebf777d266a403175d74af2bc5f34ff5872666e8:64&superpass
2aa7bde7159defd25e2bc1a4cd5f4b51e13d5579:$12tunnelme
```

With all the hashes cracked, running the last command with --show at the end will return all the newly discovered passwords.

```
(kali@testbox)-[~/Desktop]
$ hashcat -m 100 -a 7 hashes -1 ?s?d ?1?1?1 wordlist -O --show
2aa7bde7159defd25e2bc1a4cd5f4b51e13d5579:$12tunnelme
fdc01cfa3b97b2e6820be41a9f4a0d0a29dca007:donthackpls2%7
c8b347a32facbe02c2688c304b18e03eb25fcddf:uncrackable$34
ebf777d266a403175d74af2bc5f34ff5872666e8:64&superpass
f045e5e1b7195bc19bfd3026c044d93bb5048018:$98hackerman
0cc2497792b0fb6344dd03f0dd8fa27a759304cf:ursogood@67
```

Comparing these hashes with the original usernames will reveal all the SSH credentials for this network:

```
arthur      :$12tunnelme
dutch       :donthackpls2%7
trelawney   :uncrackable$34
john        :64&superpass
morgan      :$98hackerman
micah       :ursogood@67
```

Mapping the Network

With all the SSH credentials discovered, it's time to document how all the machines on this network connect to each other. An easy way to keep track of all this is to create a table that documents each machine's network interface and the assigned IP address. It's also a good idea to draw out a network map on paper if that helps with grasping the layout. A completed example table of all the addresses in this lab will be at the bottom of this section.

The first target IP address and existence of an SSH server is already known thanks to the previous NMAP scan. Trying every single set of credentials on each server will reveal what credentials match up with what machine. In the case of this walkthrough, the provided credentials happen to be in the same order as the machines are on the network.

Attempting to login to the first target is as simple as using the ssh command followed by a username and IP address. For this first box, the login is “**arthur:\$12tunnelme**”. Type yes and enter to accept the authenticity question on all of these SSH connections.

```
(kali@testbox)~$ ssh arthur@192.168.222.128
The authenticity of host '192.168.222.128 (192.168.222.128)' can't be established.
ED25519 key fingerprint is: SHA256:hLOXpyPpmG7XNZV63Q6w3vw5imQ+xCrTqRL+zwKg+vg
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.222.128' (ED25519) to the list of known hosts.
arthur@192.168.222.128's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-14-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Apr 24 16:24:16 2026 from 192.168.222.129
arthur@gateway:~$
```

The first machine on this network has been successfully hacked. The next priority from here is figuring out the IP address of the next machine to hop to. The first step to do this is using the ip a command to see what network this machine is on.

```
arthur@gateway:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:0f:73:27 brd ff:ff:ff:ff:ff:ff
    inet 192.168.222.128/24 brd 192.168.222.255 scope global dynamic noprefixroute ens33
        valid_lft 1202sec preferred_lft 1202sec
    inet6 fe80::20c:29ff:fe0f:7327/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:0f:73:31 brd ff:ff:ff:ff:ff:ff
    altnamename enp2s2
    altnamename enx00c290f7331
    inet 192.168.157.128/24 brd 192.168.157.255 scope global dynamic noprefixroute ens34
        valid_lft 1202sec preferred_lft 1202sec
    inet6 fe80::739:e372:8e41:9505/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

The .222 network was already used to gain access to this machine, so the address to pay attention to is 192.168.157.128. Now that the target network is known, it needs to be scanned in order to identify any other machines on it. This machine is just a regular Linux box, so it doesn't have tools like the previously used netdiscover or nmap. Without these utilities, the only way to scan this network is with a ping sweep. This can be accomplished with the following built-in Linux command.

```
arthur@gateway:~$ seq 1 254 | xargs -P 100 -I{} ping -c2 192.168.157.{ } | grep 'bytes from'
64 bytes from 192.168.157.128: icmp_seq=1 ttl=64 time=0.018 ms
64 bytes from 192.168.157.129: icmp_seq=1 ttl=64 time=0.516 ms
64 bytes from 192.168.157.128: icmp_seq=2 ttl=64 time=0.021 ms
64 bytes from 192.168.157.129: icmp_seq=2 ttl=64 time=0.129 ms
```

This returns successful pings on two devices. Since .128 is the address of the current machine, that means 192.168.157.129 is the IP address of the next target machine.

Now this next machine can be accessed using ssh

(ssh dutch@192.168.157.129, donthackpls2%7)

```
arthur@gateway:~$ ssh dutch@192.168.157.129
The authenticity of host '192.168.157.129 (192.168.157.129)' can't be established.
ED25519 key fingerprint is: SHA256:4A1002m2EKAhOapuHwgFwV6yZJFTQ33/f+B+XJ3dqE
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.157.129' (ED25519) to the list of known hosts.
dutch@192.168.157.129's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr 25 03:37:58 AM UTC 2026

System load: 0.08           Memory usage: 29%          Processes:   215
Usage of /: 44.3% of 9.75GB Swap usage:   0%          Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

dutch@second-hop:~$
```


This process will be repeated for the rest of the target machines until it doesn't work anymore.

Run ip a on the second target.

```
dutch@second-hop:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:77:5d:96 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.157.129/24 metric 100 brd 192.168.157.255 scope global dynamic ens33
        valid_lft 1307sec preferred_lft 1307sec
    inet6 fe80::20c:29ff:fe77:5d96/64 scope link
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:77:5d:a0 brd ff:ff:ff:ff:ff:ff
    altname enp2s2
    inet 192.168.118.128/24 metric 100 brd 192.168.118.255 scope global dynamic ens34
        valid_lft 1307sec preferred_lft 1307sec
    inet6 fe80::20c:29ff:fe77:5da0/64 scope link
        valid_lft forever preferred_lft forever
```

Ping sweep on the next identified network. (192.168.118.0)

```
dutch@second-hop:~$ seq 1 254 | xargs -P 100 -I{} ping -c2 192.168.118.{} | grep 'bytes from'
64 bytes from 192.168.118.128: icmp_seq=1 ttl=64 time=0.018 ms
64 bytes from 192.168.118.129: icmp_seq=1 ttl=64 time=0.246 ms
64 bytes from 192.168.118.128: icmp_seq=2 ttl=64 time=0.024 ms
64 bytes from 192.168.118.129: icmp_seq=2 ttl=64 time=0.220 ms
```

SSH into the newly discovered target machine.

(ssh trelawney@192.168.118.129, uncrackable\$34)

```
dutch@second-hop:~$ ssh trelawney@192.168.118.129
The authenticity of host '192.168.118.129 (192.168.118.129)' can't be established.
ED25519 key fingerprint is SHA256:QKtLRLfuPHGEY6n1vDTAqcFeRwoUyc4JWHcru6sP4vs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.118.129' (ED25519) to the list of known hosts.
trelawney@192.168.118.129's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr 25 03:49:45 AM UTC 2026

System load: 0.0          Memory usage: 28%        Processes:      216
Usage of /:  44.3% of 9.75GB Swap usage:  0%        Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

trelawney@third-hop:~$
```

Run ip a on the third target.

```
trelawney@third-hop:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:ae:81:1a brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.118.129/24 metric 100 brd 192.168.118.255 scope global dynamic ens33
        valid_lft 1721sec preferred_lft 1721sec
    inet6 fe80::20c:29ff:feae:811a/64 scope link
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:ae:81:24 brd ff:ff:ff:ff:ff:ff
    altname enp2s2
    inet 192.168.54.129/24 metric 100 brd 192.168.54.255 scope global dynamic ens34
        valid_lft 1724sec preferred_lft 1724sec
    inet6 fe80::20c:29ff:feae:8124/64 scope link
        valid_lft forever preferred_lft forever
```

Ping sweep on the next identified network. (192.168.54.129)

```
trelawney@third-hop:~$ seq 1 254 | xargs -P 100 -I{} ping -c2 192.168.54.{} | grep 'bytes from'
64 bytes from 192.168.54.128: icmp_seq=1 ttl=64 time=0.355 ms
64 bytes from 192.168.54.129: icmp_seq=1 ttl=64 time=0.014 ms
64 bytes from 192.168.54.128: icmp_seq=2 ttl=64 time=0.198 ms
64 bytes from 192.168.54.129: icmp_seq=2 ttl=64 time=0.019 ms
```

SSH into the newly discovered target machine

(ssh john@192.168.54.128, 64&superpass)

```
trelawney@third-hop:~$ ssh john@192.168.54.128
The authenticity of host '192.168.54.128 (192.168.54.128)' can't be established.
ED25519 key fingerprint is SHA256:248DuwneI6y5Sck1B9MyFir6sqwJEuYnSoXNAGw1mBo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.54.128' (ED25519) to the list of known hosts.
john@192.168.54.128's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr 25 03:55:17 AM UTC 2026

System load: 0.0           Memory usage: 28%        Processes:   213
Usage of /:  44.3% of 9.75GB Swap usage:   0%        Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

john@fourth-hop:~$
```

Run ip a on the fourth target.

```
john@fourth-hop:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:e3:a6:7c brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.54.128/24 metric 100 brd 192.168.54.255 scope global dynamic ens33
        valid_lft 1174sec preferred_lft 1174sec
    inet6 fe80::20c:29ff:fee3:a67c/64 scope link
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:e3:a6:86 brd ff:ff:ff:ff:ff:ff
    altname enp2s2
    inet 192.168.4.128/24 metric 100 brd 192.168.4.255 scope global dynamic ens34
        valid_lft 1169sec preferred_lft 1169sec
    inet6 fe80::20c:29ff:fee3:a686/64 scope link
        valid_lft forever preferred_lft forever
john@fourth-hop:~$
```

Ping sweep on the next identified network. (192.168.4.0)

```
john@fourth-hop:~$ seq 1 254 | xargs -P 100 -I{} ping -c2 192.168.4.{} | grep 'bytes from'
64 bytes from 192.168.4.128: icmp_seq=1 ttl=64 time=0.018 ms
64 bytes from 192.168.4.129: icmp_seq=1 ttl=64 time=0.268 ms
64 bytes from 192.168.4.128: icmp_seq=2 ttl=64 time=0.040 ms
64 bytes from 192.168.4.129: icmp_seq=2 ttl=64 time=0.204 ms
```

SSH into the newly discovered target machine

(ssh morgan@192.168.4.129, \$98hackerman)

```
john@fourth-hop:~$ ssh morgan@192.168.4.129
The authenticity of host '192.168.4.129 (192.168.4.129)' can't be established.
ED25519 key fingerprint is SHA256:6MWDp8TjVjVqU5F+ZsfCsW/yM0FYh9fZ9STNGCpBTS4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.4.129' (ED25519) to the list of known hosts.
morgan@192.168.4.129's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr 25 04:02:43 AM UTC 2026

System load: 0.24          Memory usage: 29%        Processes:      217
Usage of /:  44.3% of 9.75GB Swap usage:   0%         Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

morgan@fifth-hop:~$
```


Run ip a on fifth target.

```
morgan@fifth-hop:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5f:39:0d brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.4.129/24 metric 100 brd 192.168.4.255 scope global dynamic ens33
        valid_lft 926sec preferred_lft 926sec
    inet6 fe80::20c:29ff:fe5f:390d/64 scope link
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5f:39:17 brd ff:ff:ff:ff:ff:ff
    altname enp2s2
    inet 192.168.244.128/24 metric 100 brd 192.168.244.255 scope global dynamic ens34
        valid_lft 924sec preferred_lft 924sec
    inet6 fe80::20c:29ff:fe5f:3917/64 scope link
        valid_lft forever preferred_lft forever
```

Ping sweep on the next identified network. (192.168.244.0)

```
morgan@fifth-hop:~$ seq 1 254 | xargs -P 100 -I{} ping -c2 192.168.244.{} | grep 'bytes from'
64 bytes from 192.168.244.128: icmp_seq=1 ttl=64 time=0.018 ms
64 bytes from 192.168.244.130: icmp_seq=1 ttl=64 time=0.301 ms
64 bytes from 192.168.244.128: icmp_seq=2 ttl=64 time=0.022 ms
64 bytes from 192.168.244.130: icmp_seq=2 ttl=64 time=0.162 ms
```

Attempting to SSH into the next machine will fail, suggesting this next machine is not an SSH server. This is a likely indicator that this is the final target machine.

```
morgan@fifth-hop:~$ ssh micah@192.168.244.130
ssh: connect to host 192.168.244.130 port 22: Connection refused
```

Below is a fully filled out IP table for the network. At this point, the entire network has been mapped out.

	Box	IP Address
	Kali (eth1)	192.168.222.130
Box 1 inside	gateway (eth0)	192.168.222.128
Box 1 outside	gateway (eth1)	192.168.157.128
Box 2 inside	second-hop (eth0)	192.168.157.129
Box 2 outside	second-hop (eth1)	192.168.118.128
Box 3 inside	third-hop (eth0)	192.168.118.129
Box 3 outside	third-hop (eth1)	192.168.54.129
Box 4 inside	fourth-hop (eth0)	192.168.54.128

Box 4 outside	fourth-hop (eth1)	192.168.4.128
Box 5 inside	fifth-hop (eth0)	192.168.4.129
Box 5 outside	fifth-hop (eth1)	192.168.244.128
Box 6 inside	final-destination (eth0)	192.168.244.130

SSH Tunneling

With the entire network documented, the only thing standing in the way of accessing the last machine is building out the SSH tunnels with listeners. The easiest way to keep track of all the different interfaces throughout this process is to refer to them as “inside” and “outside” interfaces. This is documented in the left-most column of the above table. These do not include Kali as it isn’t necessary to use Kali’s IP to create any of the SSH listeners.

An SSH listener is created by using the command `ssh -L`. A port for the listener must be assigned at the beginning of the IP address, and the ssh port (22) at the end of it. For the purpose of keeping things as simple as possible, each listener will be assigned an incrementing number. (EX: 1111, 2222, 3333, etc.)

Create a new terminal window for every single SSH listener.

Do not close any SSH listeners unless a mistake was made in configuration.

The first listener needs to be set up from the inside of box 2 to the inside of box 1.

```
ssh -L 1111:192.168.157.129:22 arthur@192.168.222.128
```

```
(kali@testbox)-[~/Desktop]
$ ssh -L 1111:192.168.157.129:22 arthur@192.168.222.128
arthur@192.168.222.128's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-14-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Apr 24 22:44:51 2026 from 192.168.222.130
arthur@gateway:~$
```


The second listener needs to be set up from the outside of box 2 to the loopback address.

Each subsequent listener must include -p accompanied by the port assigned to the previous tunnel.

```
ssh -L 2222:192.168.118.128:22 dutch@127.0.0.1 -p 1111
```

```
(kali@testbox)-[~/Desktop]
$ ssh -L 2222:192.168.118.128:22 dutch@127.0.0.1 -p 1111
The authenticity of host '[127.0.0.1]:1111 ([127.0.0.1]:1111)' can't be established.
ED25519 key fingerprint is: SHA256:4A1002m2EkAh0apuHWgFwV6yZJFToQ3J/f+B+XJ3dqE
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[127.0.0.1]:1111' (ED25519) to the list of known hosts.
dutch@127.0.0.1's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr 25 04:40:29 AM UTC 2026

System load: 0.08          Memory usage: 29%    Processes:      218
Usage of /:  44.3% of 9.75GB Swap usage:   0%      Users logged in: 1

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sat Apr 25 03:37:59 2026 from 192.168.157.128
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

dutch@second-hop:~$
```

The third listener needs to be set up from the inside of box 3 to the loopback address.

```
ssh -L 3333:192.168.118.129:22 dutch@127.0.0.1 -p 2222
```

```
(kali@textbox)-[~/Desktop]
$ ssh -L 3333:192.168.118.129:22 dutch@127.0.0.1 -p 2222
The authenticity of host '[127.0.0.1]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is: SHA256:4A1002m2EkAh0apuHWgFwV6yZJFTQ3J/f+B+XJ3dqE
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[127.0.0.1]:2222' (ED25519) to the list of known hosts.
dutch@127.0.0.1's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr 25 04:44:20 AM UTC 2026

System load: 0.03          Memory usage: 30%      Processes:      219
Usage of /:  44.3% of 9.75GB Swap usage:   0%        Users logged in: 1

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Sat Apr 25 04:40:29 2026 from 192.168.157.128
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

dutch@second-hop:~$
```

The fourth listener needs to be set up from the inside of box 4 to the loopback address.

```
ssh -L 4444:192.168.54.128:22 trelawney@127.0.0.1 -p 3333
```

```
(kali@textbox)-[~/Desktop]
$ ssh -L 4444:192.168.54.128:22 trelawney@127.0.0.1 -p 3333
The authenticity of host '[127.0.0.1]:3333 ([127.0.0.1]:3333)' can't be established.
ED25519 key fingerprint is: SHA256:QKtLRLfuPHGEY6n1vDTAqcFeRwoUyc4JWHcru6sP4vs
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[127.0.0.1]:3333' (ED25519) to the list of known hosts.
trelawney@127.0.0.1's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr 25 04:49:24 AM UTC 2026

System load: 0.0          Memory usage: 30%      Processes:      219
Usage of /:  44.3% of 9.75GB Swap usage:   0%        Users logged in: 1

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Sat Apr 25 03:49:45 2026 from 192.168.118.128
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

trelawney@third-hop:~$
```

The fifth listener needs to be set up from the inside of box 5 to the loopback address

Ssh -L 5555:192.168.4.129:22 john@127.0.0.1 -p 4444

```
(kali@testbox)-[~/Desktop]
$ ssh -L 5555:192.168.4.129:22 john@127.0.0.1 -p 4444
The authenticity of host '[127.0.0.1]:4444 ([127.0.0.1]:4444)' can't be established.
ED25519 key fingerprint is: SHA256:248DuwneI6y5Sck1B9MyFir6sqwJEuYnSoXNAGw1mBo
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[127.0.0.1]:4444' (ED25519) to the list of known hosts.
john@127.0.0.1's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr 25 04:54:05 AM UTC 2026

System load: 0.0          Memory usage: 30%    Processes:      217
Usage of /:  44.3% of 9.75GB Swap usage:   0%    Users logged in: 1

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Sat Apr 25 03:55:18 2026 from 192.168.54.129
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

john@fourth-hop:~$
```

The sixth listener needs to be set up from the inside of box 6 to the loopback address

This listener needs to go through port 80 rather than 22, as it is the last listener before the final machine where it will be used to access an http webpage.

The port is also incremented to 5556 instead of 6666, as the browser blocks web browsing traffic on port 6666.

ssh -L 5556:192.168.244.130:80 morgan@127.0.0.1 -p 5555

```
(kali@testbox)-[~/Desktop]
$ ssh -L 5556:192.168.244.130:80 morgan@127.0.0.1 -p 5555
morgan@127.0.0.1's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr 25 05:08:04 AM UTC 2026

System load: 0.0          Memory usage: 31%    Processes:      217
Usage of /:  44.3% of 9.75GB Swap usage:   0%    Users logged in: 1

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

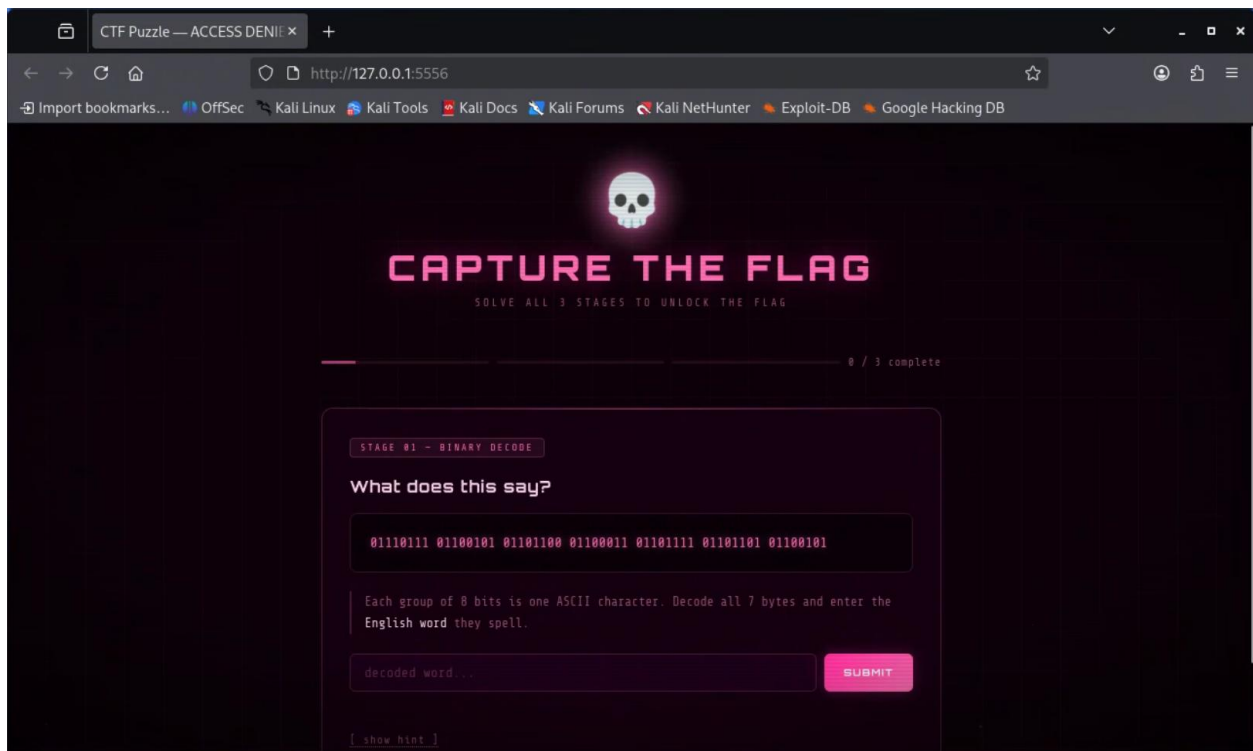
Last login: Sat Apr 25 05:06:07 2026 from 192.168.4.128
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

morgan@fifth-hop:~$
```

The success of the tunnels can be tested by opening the web page on the final target machine. This can be done by using the firefox command along with the loopback address and the most previous listener port.

```
(kali@testbox)-[~/Desktop]  
$ firefox http://127.0.0.1:5556
```

This will open the webpage in firefox.



At this point, full access has been gained to the entire network. It would be possible to go through this final web page and acquire the final flag to complete the challenge.

However, more can be done with SSH tunneling. A dynamic listener can tie together all the previously created tunnels, excluding the last one. This would create a direct connection from the first Kali box to the final target machine. Using this with proxychains would allow direct exploitation to the final machine with tools like NMAP, dirb, and any other tool that can be made to work with proxychains.

Dynamic Listener

A dynamic listener can be created using the argument `-D` rather than the previously used `-L`. A unique port must be assigned to the listener, and it points to the loopback at the second-most recent listener with the most recent username.

```
ssh -D 10001 morgan@127.0.0.1 -p 5555
```

```
(kali@testbox)-[~/Desktop]
$ ssh -D 10001 morgan@127.0.0.1 -p 5555
morgan@127.0.0.1's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr 25 05:35:30 AM UTC 2026

System load: 0.0          Memory usage: 30%       Processes:      218
Usage of /:  44.3% of 9.75GB Swap usage:   0%       Users logged in: 1

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Sat Apr 25 05:08:05 2026 from 192.168.4.128
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

morgan@fifth-hop:~$
```

After creating the dynamic listener, the proxychains configuration file needs to be updated to use it. This can be edited with `sudo nano /etc/proxychains4.conf`.

```
(kali@testbox)-[~/Desktop]
$ sudo nano /etc/proxychains4.conf
```

The following changes need to be made within the conf file:

Comment out `proxy_dns`.

```
# method 1. this uses the proxychains4 style method to do remote dns:
# a thread is spawned that serves DNS requests and hands down an ip
# assigned from an internal list (via remote_dns_subnet).
# this is the easiest (setup-wise) and fastest method, however on
# systems with buggy libcs and very complex software like webbrowsers
# this might not work and/or cause crashes.
# proxy_dns
```

```
# Some timeouts in milliseconds
tcp_read_time_out 1500
tcp_connect_time_out 800
```

```
##
[ProxyList]
# add proxy here ...
# meanwhile
# defaults set to "tor"
# socks4          127.0.0.1 9050
socks4            127.0.0.1 10001
```

```
proxychains dirb http://192.168.244.130
```

[illegible]

NMAP can also be used to check for services, although using it through proxychains results in everything appearing as filtered.

```
nmap -sT -v -Pn -n -p 80 192.168.244.130
```

```
(kali@testbox)-[~/Desktop]
$ nmap -sT -v -Pn -n -p 80 192.168.244.130
Host discovery disabled (-Pn). All addresses will be marked 'up' and scan times may be slower.
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-25 00:48 -0500
Initiating Connect Scan at 00:48
Scanning 192.168.244.130 [1 port]
Completed Connect Scan at 00:48, 2.00s elapsed (1 total ports)
Nmap scan report for 192.168.244.130
Host is up.

PORT      STATE      SERVICE
80/tcp    filtered  http

Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 2.03 seconds
```

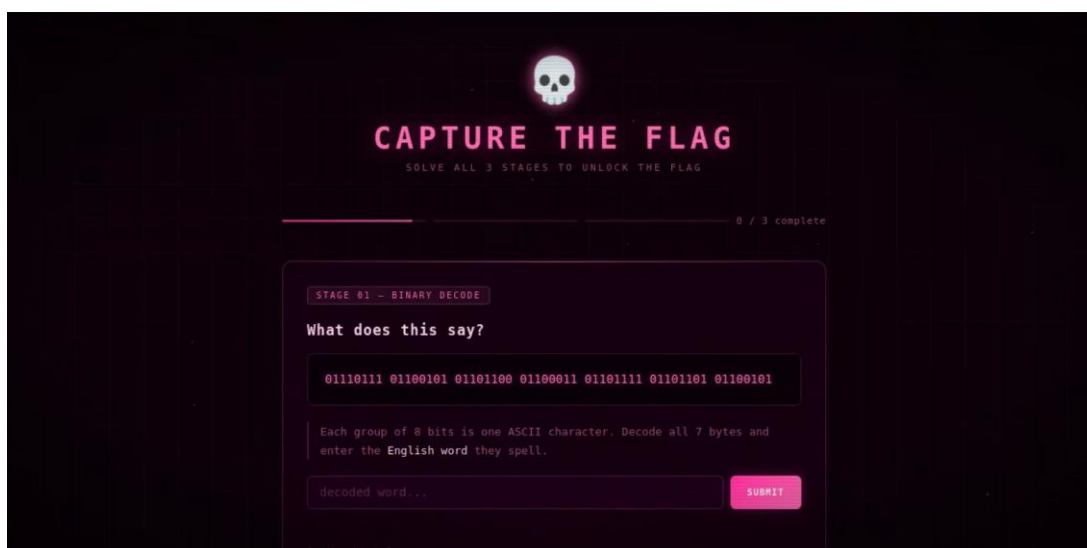
With all the tunnels set up, all that's left is to grab the final flag.

The Flag

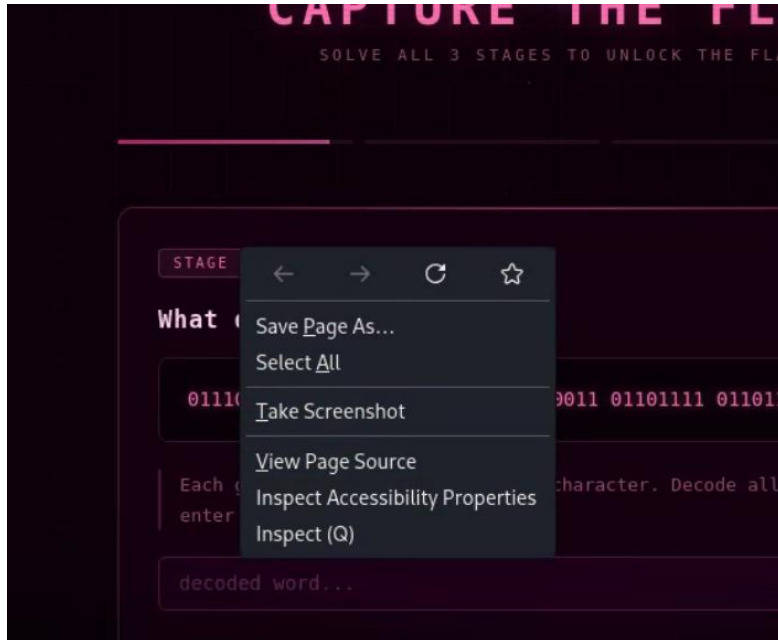
Just like the other tools, proxychains can be used with firefox to access the previously visited web page.

```
(kali@testbox)-[~/Desktop]
$ proxychains firefox http://192.168.244.130
```

Once again, this pulls up the capture the flag web page.



There are two ways to go about extracting this final flag. Solving all three of the provided challenges would result in the flag being displayed by the webpage. However, this page is using client-side JavaScript to deliver this challenge. This makes it possible to once again view the source code.



Inspection of the `<script>` section reveals a win section that contains the final flag.

```
<!-- Win -->
<div class="stage" id="stage-win">
  <span class="win-icon">&#x1F3C6;</span>
  <h2 class="win-title">ACCESS GRANTED</h2>
  <p class="win-sub">Binary decoded. Cipher cracked. Riddle solved.<br>Here is your flag:</p>
  <div class="flag-box">flag{th4t_w4s_3asy}</div>
</div>
```

The final flag for this challenge is: “flag{th4t_w4s_3asy}”